

Evaluation of a targeted behavioural sleep intervention for student-athletes

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RATIONALE Previous research has identified student-athletes as a group that exhibits poor sleep patterns due to academic and sport-related risk factors¹. Although many sleep interventions for athletes adopt a behavioural approach, there is evidence to suggest a lack of behaviour change theory underpinning their development². Therefore, this pilot study aimed to assess the effectiveness of a targeted intervention to modify sleep behaviour in student-athletes.

METHODS The study received institutional ethical approval (ETHICS2023-83) and was conducted in accordance with the principles of the Declaration of Helsinki.

A behavioural diagnosis was conducted using semi-structured interviews, which were analysed through reflexive thematic analysis. The identified themes were then mapped onto the COM-B behavioural model³. Three COM-B components were identified as feasible barriers to address in the intervention (Figure 1). Further details on this process can be accessed via the QR code at the poster end.

The intervention was designed with two key parts: a 40-minute in-person interactive workshop, and personalised visual feedback on participants' baseline sleep data. These components incorporated nine distinct behaviour change techniques — considered to be essential, active, and irreducible elements for modifying behaviour — from the Behaviour Change Technique Taxonomy⁴ (Figure 2).

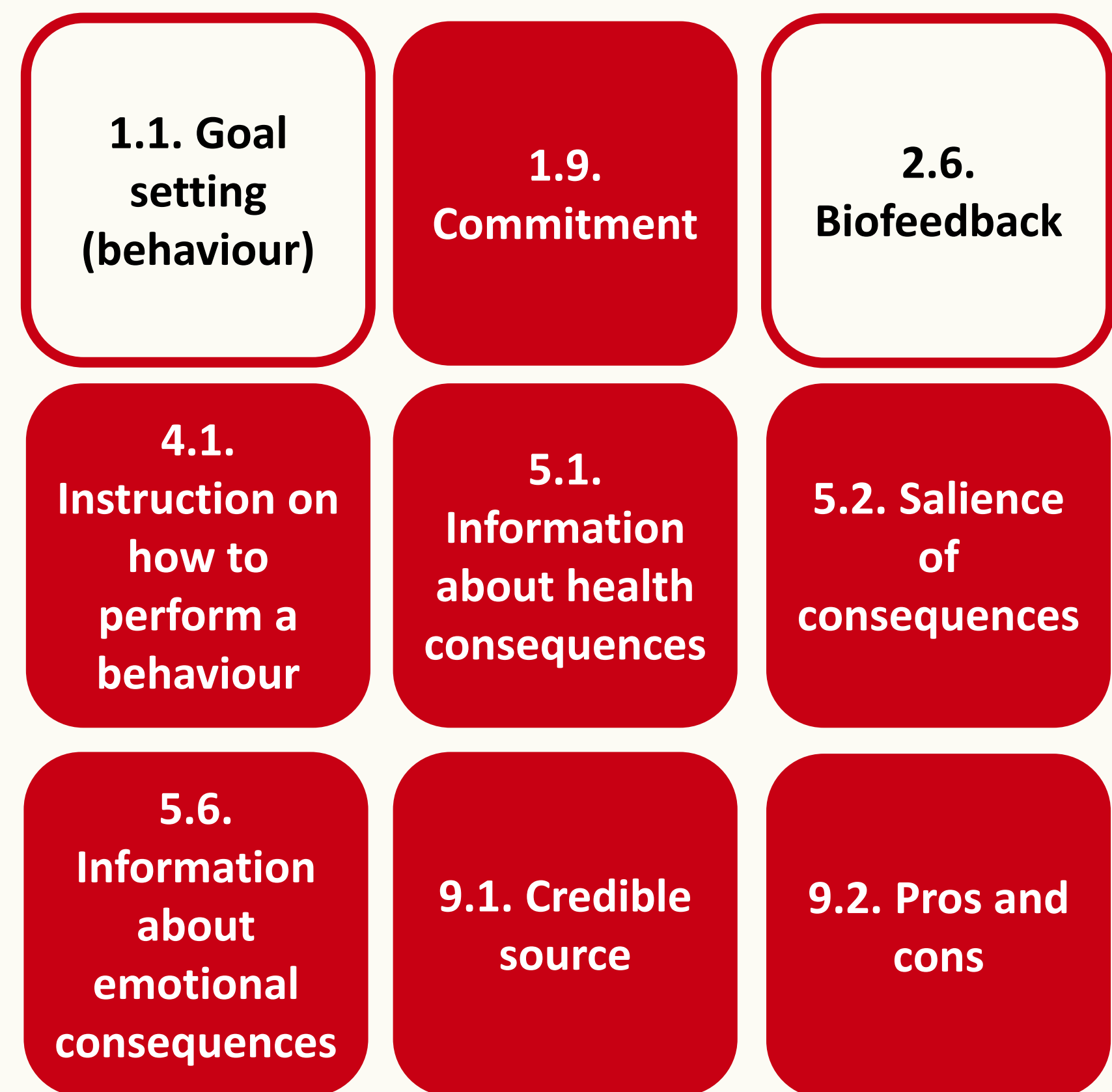


Figure 2. Behaviour change techniques incorporated into the workshop (red) or feedback (clear) intervention parts.

Fifteen male Rugby Union student-athletes (mean age: 19 ± 1 year) competing at a regional level completed the protocol. Participants underwent one week of wrist-worn actigraphy (GENEActiv, Activinsights) accompanied by a sleep diary, followed by attendance at the workshop upon returning. One week later, they received their sleep feedback via email. Five weeks after the initial data collection, participants completed a second week of actigraphy and sleep diary recording (Figure 3).

Actigraphy data were processed and analysed in R using the GGIR open-source package⁵. Descriptive analysis and paired Wilcoxon signed-rank tests were conducted to compare outcomes between baseline and follow-up actigraphy.

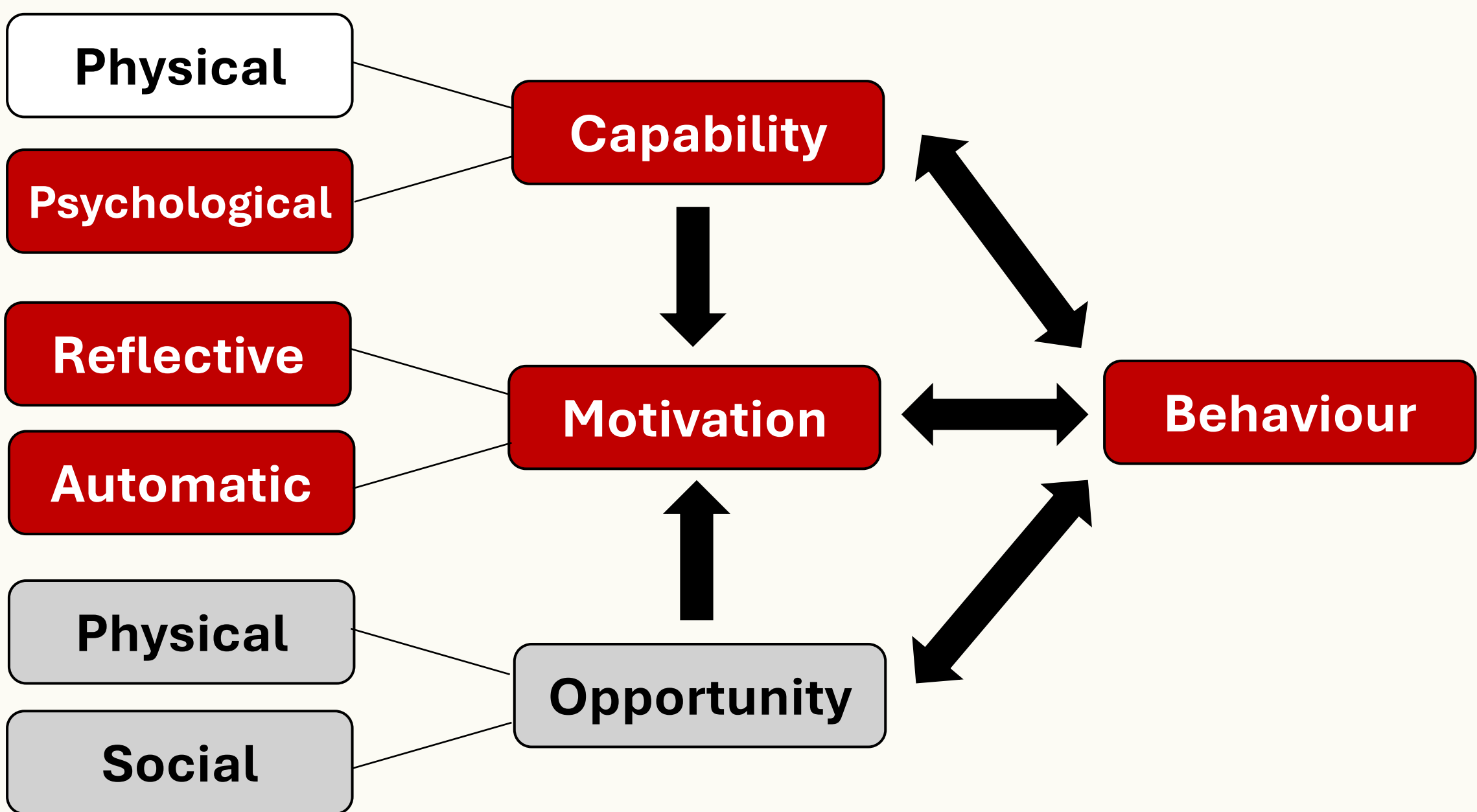


Figure 1. Behavioural diagnosis using COM-B model. Red: identified as barrier and included in intervention design; Grey: identified as barrier but not feasible to address; White: not identified as barrier.

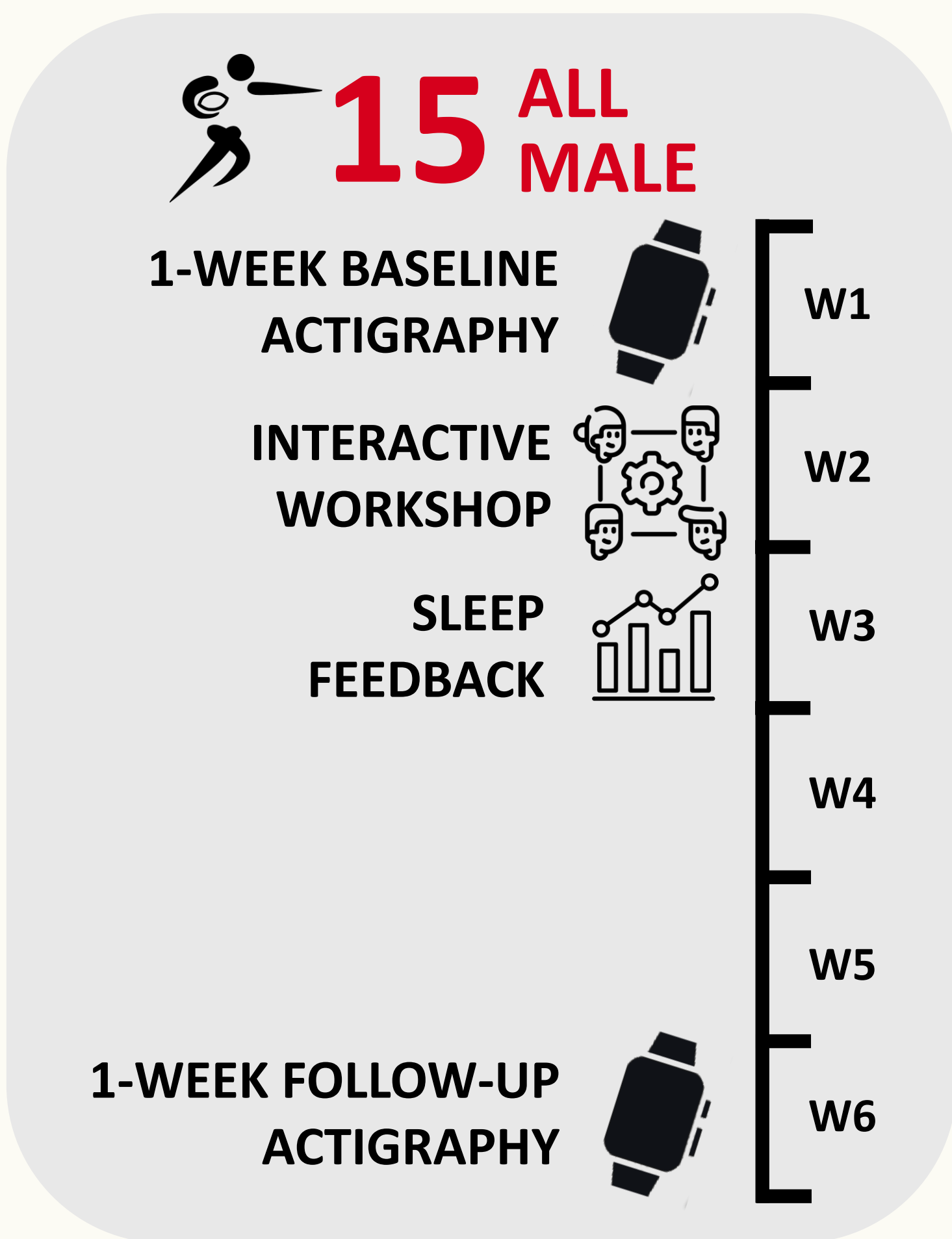


Figure 3. Schematic of study protocol.

RESULTS Baseline actigraphy data revealed short nocturnal sleep durations and inconsistent sleep/wake timings. No statistically significant differences were observed between follow-up and baseline for any sleep parameter (all $p > .05$; Table 1).

DISCUSSION The findings corroborate previous research indicating suboptimal sleep parameters in student-athletes. Although the intervention did not significantly alter any sleep parameters, it may help protect against the deterioration in sleep often observed over the course of an academic semester. An intervention incorporating the COM-B components identified but not included in this pilot study for feasibility reasons (e.g., addressing physical opportunity through scheduling changes) may offer a more effective approach to promoting sleep behaviour change in student-athletes.

REFERENCES

1. Kroshus et al. (2019). *Br J Sports Med*, 53(12), 731-736.
2. Wilson et al. (2024). *Behav Sleep Med*, 1-23 [advance online publication].
3. Michie et al. (2011). *Implement Sci*, 6(1):42.
4. Michie et al. (2013). *Ann Behav Med*, 46(1), 81-95.
5. Migueles et al. (2019). *J Meas Phys Behav*, 2(3), 188-196.

Additional resources and information



Table 1. Sleep parameters at baseline and follow-up

Variable	Baseline	Post-intervention	Effect size (95% CI)
Sleep period (hr)	7.51 (6.98 to 8.03)	7.78 (7.08 to 8.30)	0.08 (-0.50, 0.67)
Total sleep time (hr)	6.43 (5.97 to 6.99)	6.47 (5.92 to 7.17)	0.20 (-0.38, 0.73)
Total sleep time incl. naps (hr)	6.69 (6.24 to 7.08)	7.06 (6.31 to 7.42)	0.42 (-0.10, 0.88)
Wake after sleep onset (hr)	1.09 (0.87 to 1.31)	1.04 (0.93 to 1.20)	-0.15 (-0.70, 0.47)
Sleep efficiency (%)	85.9 (83.2 to 88.0)	86.5 (83.7 to 89.0)	0.19 (-0.39, 0.73)
Sleep onset (hh:mm)	01:06 (00:42 to 01:54)	01:08 (00:44 to 01:31)	-0.34 (-0.82, 0.25)
Sleep offset (hh:mm)	08:36 (08:29 to 08:53)	09:01 (08:15 to 09:09)	-0.16 (-0.70, 0.46)
Sleep Regularity Index (units)	70.5 (66.1 to 73.6)	69.5 (63.1 to 79.6)	0.06 (-0.51, 0.65)

Variables at baseline and post-intervention presented as median (interquartile range)